

SUMMARY

Senior Network Engineer with over 11 years of extensive experience in design, architecture, development, implementation, configuration, maintenance, optimization, integration and troubleshooting of enterprise networks. I have experience within both Enterprise and telecom network environments, in the private sector and the public sector, greenfield and brownfield, cloud, on-premise and hybrid

Proficiency with multiple areas of network services; including Network security, Cyber Security, cloud security, Software Defined Networking (SDWAN/ACI), Network Automation and Orchestration.

Several years of architecting and designing complex LAN/WAN/Datacenter network infrastructures, including implementation of security technologies and solutions.

Conducted performance testing and validation of automated deployments across AWS VPCs, Aruba switches, and Meraki cloud controllers.

Architecting, configuring, deployment of NextGen firewalls, routers, and switches. Routing Protocols: BGP, OSPF, EIGRP, VRF/VPNs, including Day -1, Day – 2 operations.

Performing regular code upgrades for Cisco routers, switches, Next-Generation firewalls: FortiGate firewalls, Fortimanager, FortiAnalyzer, Cisco Secure Firewall (Cisco Firepower) and Cisco Secure Firewall Management Center (Cisco Firepower Management Center), Palo Alto, Panorama/Strata Cloud Manager (SCM) management for both on premises and cloud-based solutions, integrating both physical and virtualized platforms.

Delivering timely and deliberate engagement during major system outage events, using good communication skills, good decision-making skills, and documentation skills needed to produce an effective return-to-service and an actionable root-cause-analysis to minimize/avoid future problems

CERTIFICATIONS AND TRAINING:

CCNP Enterprise
CCDP (Cisco Certified Design Professional)
Cisco Certified Specialist - Enterprise Core
Cisco Certified Specialist - Enterprise Advanced Infrastructure Implementation
Cisco Certified Specialist - Enterprise Design
Cisco Certified Specialist - Enterprise SD-WA Implementation
CCNA (Cisco Certified Network Associate)
CCNA Routing and Switching
CCDA - Cisco Certified Design Associate
Fortinet Network Security Expert (NSE4)
JNCIA-SEC: Security, Associate
JNCIS-SEC: Security, Specialist
CCNA (Security)
CompTIA Security+
CompTIA Network+

EDUCATION

Bachelor of Science in Computer Science from Makerere University
Career Training: PCAGE Career Institute – Parsippany New Jersey

PROFESSIONAL EXPERIENCE:

Securus Technologies/Aventiv Technologies LLC

Apr 2023 to Present – Carrollton TX

Sr. Network Security Engineer – FRE Teams

Designed, built, deployed, administered, and troubleshot Fortinet firewalls and other Fortinet network platforms.

Performed conversion of cisco ASA firewalls to Fortinet/FortiGate NGFW using Forti Converter.

Performed application, network, and firewall provisioning- working with Forti manager, performing changes to monitor/block/allow/filter various traffic (such as NAT security, Whitelisting of IPs, security profiles, URL filtering, security zones, firewall policies etc.).

Performed code upgrades for FortiGate firewalls and Forti switches as well as troubleshoot Fortinet platforms using tools such as Forti manager, Forti analyzer, Forti Cloud, and Forti Gates.

Integrated both physical and virtualized firewalls with Forti Manager, Forti Analyzer, Forti Authenticator and other Fortinet products as required.

Reviewed existing Cisco Meraki firewall rules and Cisco ASA rules to identify stale, infrequently used, or never used rules for a planned migration to the new Next-Generation firewall platforms.

Provided training and mentorship on network automation best practices for cross-functional engineering teams.

Integrated cloud networking solutions with AWS, enabling hybrid connectivity and secure workload migration.

Diagnosed problems and troubleshoot Meraki product line, with particular emphasis on Meraki security appliances and various network-related issues reported by customers.

Configured firewall interfaces, network configurations, access control lists and policies, static and dynamic routing, remote access (VPN functionality) and implemented best practices for next-generation firewall feature sets.

Configured and troubleshoot dynamic routing protocols OSPF, and BGP, while leveraging both on-prem and cloud technologies.

Configured and managed Aruba and Meraki wireless solutions, ensuring reliable and scalable enterprise wireless connectivity.

Designed end-to-end observability automation pipelines using Terraform, Ansible, to capture network metrics and auto-scale resources.

Implemented and configured robust FortiGate SDWAN 6.x and 7.x solutions for intelligent traffic management across multiple WAN connections, prioritizing applications based on business needs while incorporating advanced security features, essentially providing a secure and optimized network for distributed enterprises.

Monitored the enterprise network infrastructure, performing interventions, responding to distress, and investigating breaches to prevent them from recurring, and ensuring the security of enterprise network and digital assets.

Ensured high availability and resiliency of network services by automating failover, monitoring, and scaling processes.

Ensured the smooth operation of Cisco Nexus OS product suite and handled any challenges that arose.

Identified problems before they happen – owning incident response and escalations.

Configured and implemented firewall security policies and network segmentation in Palo Alto NGFW firewalls and Microsoft Azure cloud.

Troubleshoot network issues caused by security tools to enable business functionality while protecting the client's network infrastructure and information.

Owned enterprise-wide incident response, triaging and escalations.

Collaborated with other support team members to fix network outages, misconfigurations, and firewall/network related issues on clients' devices.

Worked with other client teams and management to diagnose and resolve critical, escalated issues.

Analysed and leveraged Wireshark packet captures, traffic logs from Palo alto NGFW, tcpdump and Microsoft Sentinel to aid in troubleshooting.

Granite Telecommunications

Jul 2022 to Feb 2023 Dallas TX

Senior Network/Security Engineer

Designed and implemented network monitoring solutions, configured monitoring systems, and provisioned network services for various network platforms.

Managed the network infrastructure for deployments, made decisions on how to connect all the systems and how to organize the underlying networks, as well as troubleshoot network issues with infrastructure.

Worked with FortiOS 6.x & v7.x, virtual firewalls; Forti Manager, Forti Analyzer & ADOMs.

Configured and performed analysis of network security needs, configured firewall settings, IPSEC VPNs, GRE tunnels, firewall policies, address/security groups, and ACLs.

Designed, configured, and troubleshoot IP routing and switching including BGP, MPLS, and OSPF.

Assessed, designed, and implemented modern secure networking environments on multiple vendor environments including Fortinet (FortiGate firewalls, FortiSwitches, FortiAp, FortiNAC, FortiSiem), Cisco (Catalyst, Nexus, ASR/ISR) and other technologies.

Implemented end-to-end SD-WAN automation pipelines integrating CodeShuttle with CI/CD workflows for rapid and error-free deployments.

Architected and designed complex LAN/WAN network infrastructure, including implementation of both network and security technologies.

Performed root cause analysis of critical network outage and incidents.

Handled daily BAU tasks including network diagnostics, VPN configurations, on Palo Alto and FortiGate firewall policy changes, activation and turn up of customer security equipment, and failover testing.

Integrated CloudWatch, Azure Monitor, and SolarWinds APIs to automate alerting, logging, and performance analysis.

Collaborated with cloud architects to design hybrid network topologies, automating connectivity between AWS workloads and on-premises data centers.

Partnered with cloud and security teams to design automated workflows for network provisioning, compliance checks, and incident response.

Performed PANOS and Fortigate firewall upgrades, configured HA, NAT, Security Zones, Security profiles, Security Policy creation in line with security standards.

Provided design considerations for SASE (Secure Access Service Edge) architecture, such as secure web gateway (SWG), NGFW, zero-trust network access (ZTNA), and Secure SD-WAN, thereby facilitating the ability to extend secure access and high-performance connectivity to users.

Worked with SIEM/SIM reporting tools such as Splunk Enterprise security solution and FortiSIEM solution from Fortinet.

Managed Palo Alto and Fortigate Firewalls via Panorama and Fortimanager respectively.

Wrote template/scripts as well as created and pushed device settings, security policies, routing policies onto managed Palo Alto and Fortigate Firewalls.

Bank of America

Aug 2021 to Jul 2022 – Richardson TX

Network Engineer/Automation Engineer/ (Core and Data Center Networks)

Architected, designed, built, and operated large scale network systems and services across the enterprise.

Implemented efficient, high availability, resilient networks, across the routing, switching, datacenter, Data Center, LAN, WAN, Internet Edge, Cloud Connectivity, including SD-WAN environments.

Provided continuous service improvement of the network environment, both in terms of performance and reduction in the cost of support via network automation.

Supported network deployment, configuration, maintenance, management, and ensured network lifecycle best practices.

Set up standards for network configuration, hardware/software, and architecture.

Partnered with vendors to understand their approach to new solutions, architectural standards, and frameworks.

Developed network strategies, architectures, designs, and standards across all core network infrastructure including global WAN/MAN, DNS/DHCP, data center networks, network security, access networks/remote access, Internet, and physical network infrastructure.

Troubleshoot and debugged complex networking problems.

Used Jira and Confluence to manage and document work

Developed automated playbooks for Palo Alto firewall policy management and IPSec VPN configuration, enhancing security and compliance.

Created accurate, representative, and usable network diagrams & documentation.

Designed and implemented cloud-based networking solutions in AWS, integrating automation for VPCs, subnets, and VPN connectivity.

Supported multiple datacenters, branch offices and corporate locations across the Global Network infrastructure.

Contributes to the development, implements, and enforces service and technology standards via automation across the network footprint.

Collaborated with other Network Architects to drive automation service improvements throughout the enterprise and business networks.

Developed, implemented, and optimized network automation solution designs of the highest level of complexity.

Leveraged Terraform to provision and manage cloud and on-premises network infrastructure as code, ensuring repeatability and governance.

Develop microservices, APIs, service templates that define the services that we provide to our customers using python and Django framework

Check in and review code, to our GIT source control repositories (Bitbucket). Effectively use Jira and Confluence to manage and document work

Deploy code to our development, QA, Staging and Production servers using our continuous integration and deployment – CI/CD pipeline platform

Test new services to ensure they are functioning as expected using our testing framework. Add new test cases to the test suite to continuously improve our quality assurance standards. Work in an Agile programming environment, including sprints, stand-ups, retrospectives, and planning

Deployed SolarWinds monitoring automation for proactive detection, alerting, and remediation of critical network performance issues.

Generally responsible for developing, deploying, and testing microservices, applications, user interfaces, ansible templates and resource adapters (plugins) in the Network Automation Engine platform. Writing python and ansible modules that deploy and test network fabrics via a controller. Writing and executing tests to validate the automation against known standards

Century Link/Level3 Communications

Nov 2017 to July 2021 Broomfield CO/Tulsa OK

Solutions Architect/Network Engineer III

Created detailed design artifacts, test plans; and independently developed and reviewed code and contributed to the go-live plan.

Collaborated and interacted with other Network Engineers to create high quality and defect-free solutions to meet business requirements.

Implemented changes to the application architecture.

Provided training and mentorship on network automation best practices for cross-functional engineering teams.

Engineered and maintained automated network provisioning workflows using Ansible,

Developed automation routines to manage Aruba switches, Meraki access points, and controllers, enabling zero-touch deployments and centralized control.

Expanded and extended network capabilities while also adding locations/users to the network, including L4-7 integration in Cisco ACI, migrating workloads into ACI fabric and configuring ACI multi-site with deployed fabric and creating new tenants, bridge domains, VRF's, and EPGs.

Collaborated with other team members to support and resolve complex technical issues within the Cisco Viptela SD-WAN architecture.

Deployed and troubleshoot PnP (plug and play), Zero-Touch Provisioning (ZTP), vBond, vSmart and Network Management Systems (NMS) such as vManage, SNMP, SolarWinds, and NetFlow.

Configured and troubleshoot hardware platforms such as routers, switches, and network interface cards.

Performed network discovery and prepared playbooks for FIS network migration

Designed, documented, implemented, and supported a sophisticated, and secure enterprise network environment, consisting of multiple branch offices, call centers, data centers, B2B and B2C connectivity.

Led network project related activities and participated in other IT projects with network related tasks.

Provided the third-tier support for all facets of the network such as VoIP, wireless, firewalls, load-balancers, routers, switches, MPLS, QoS, security, and other network environment issues

Engaged and escalated with third-party vendors as needed to address problems within the network environment.

Delivered timely and deliberate engagement during major system outage events, using good communication skills, good decision skills, and documentation skills needed to produce an effective return-to-service and an actionable root-cause-analysis to minimize/avoid future problems.

Developed, documented, and executed maintenance procedures to ensure system uptime and was pro-active in finding potential problems.

Maintained and managed border security systems, such as IPS/IDS systems and firewalls, install, configure, manage, troubleshoot, and fix Palo Alto Firewalls (PA-3220, PA-3250, and PA-3260).

Troubleshoot network hardware and solutions including but not limited to Cisco routers, switches, VPNs, firewalls, and F5 load-balancers.

Understood the network load balancing architecture, Load balancing traffic patterns (Cisco, F5, Citrix) and application requirements.

Configuring, maintaining, and optimizing cisco Firepower firewalls for optimal performance and security.

Implementing and managing security policies, access controls, and intrusion prevention systems.

Utilizing threat intelligence feeds to identify and mitigate security risks.

Evaluated the need for any new load balancer reconfigurations and made suggestions for implementation as and when required.

Managed, administered, and troubleshooted the F5 load balancer environment daily.

Ensured that systems and environment were secure and that all industry and regulatory best practice steps were taken to protect the network, hardware, software, and data from unauthorized access.

Designed, documented, configured, and installed new technology into the enterprise network environment.

eBay

May 2017 to Nov 2017 - San Jose CA

NOC Network Engineer L2/Lead Consultant

Planned, forecasted, implemented, and identified resource requirements for network systems.

Developed, maintained, and implemented security policies, specifications, and standards.

Configured, administered, and troubleshooted firewall problems in a large enterprise involving complex network application flows between multiple hosts spanning multiple firewalls/security zones and different geographic locations on platforms such as UTMs Cisco MX and Juniper SRX and other firewall services such as IPsec/SSLVPNs, Application/Web filtering and IDS/IPS systems.

Implemented Cisco ASA series Firewalls, ISE, and ACS security solutions.

Performed day-to-day engineering and operations tasks on Cisco Identity Service Engine (ISE) and Access Control System (ACS) including installations, upgrades, configuration, and troubleshooting.

Evaluated new products, performed network problem resolution, and assisted in the development and documentation of technical standards (SOPs) and interface applications.

Drove initiatives to adopt network-as-code principles, modernizing infrastructure management and reducing operational risks.

Utilized monitoring tools to provide proactive and reactive network support for eBay Inc.

Diagnosed issues with routers and routing protocols (OSPF, BGP), switches, layer two protocols and wireless uplinks.

Documented, escalated, and collaborated with higher tier engineers/other resolver teams.

Performed network changes to resolve issues and accommodate new requirements.

Performed required maintenance to contain, eradicate, and recover from network outages and infrastructure security incidents.

Responded to network alarms and took appropriate action to classify, prioritize, and resolve network and system issues.

Provided critical service outage notification and escalated issues for timely resolution.

Notified appropriate company departments and personnel as appropriate about ongoing infrastructure issues impacting business operations.

Worked with service providers to resolve circuit problems.

Assured IP allocation for dynamic and static assignments.

Provided statistical analysis of network links for capacity and trending plans.

Vendor handling and interaction by partnering with the broader eBay Network/Infrastructure organization in the areas of network engineering, configurations standards, and tools.

Nike

Nov 2015 to Jan 2017 – Beaverton OR

Senior Network Engineer

Provided ongoing operations and infrastructure support for over 1000 Nike Retail stores globally.

Worked with the Nike Retail Network Infrastructure team, external service providers as well as other internal teams such as the Mobility Engineering, Operations Intelligence Center/OIC, Security Operations Center/SOC, Global Servers team, Service Desk, Applications team to ensure that the Nike Global Network Infrastructure was working as designed.

Developed and maintained efficient Nike network communications systems.

Tested, evaluated, and troubleshooted network systems running Cisco (IOS), Juniper (JUNOS OS), as well as protocols EIGRP, OSPF, BGP, MPLS/VPN, and IPSEC to eliminate problems and suggest/make improvements.

Monitored the Nike Global Enterprise network systems and devices using SolarWinds, System Center Operations Manager (SCOM), and EM7/Science Logic

Optimized the capability of the IT infrastructure and services to deliver cost effective and sustained level of Network and application availability.

Installed hardware and software taking proactive steps to ensure success.

Monitored Aruba and Cisco Wireless Access points (WAPs) using Aruba airwave and Cisco Prime Infrastructure.

Worked with Nike Retail stores across the world to configure, support and troubleshoot the Nike Enterprise network Infrastructure

Worked with the Mobility Engineering team to facilitate the growth of mobile capabilities within Nike Retail Infrastructure and collaborated with other IT groups to ensure that computing infrastructures such as Network, Desktop, and Server systems can support the Mobility demand to ensure product availability.

Expanded current technologies and implemented Mobile Device Management (MDM) using MobileIron/Nike Retail MDM, Air-Watch Enterprise Mobility Management, Bomgar, Avalanche as well as other mobile solutions for Nike Retail.

Configured and performed mobile device imaging for IOS based devices (Apple handheld devices/iPods/iPads) as well as HHTs and mobile printers.

Monitored and received trouble tickets using Clarify and Service Now ticketing systems.

Wrote procedures (KBs, Runbooks) to respond to results so system issues are brought to resolution.

Built and managed the configuration of production systems including the management of continuous integration and deployment methodology for server-based technologies.

Maintained and managed the AWS infrastructure.

Ensured necessary system security by using best in class cloud security solutions, troubleshoot systems and solved problems across all platform and application domains.

General Electric

Jun 2014 – Oct 2015 - Minden, NV

IT Engineer/Information Systems On-Site Coordinator

Responsible for the Computer, Storage and Network provisioning through Configuring and deploying network equipment/hardware in Datacenter; configure HP and Cisco Switches/Layer 3 Switches/Routers; Design, deploy and support LANs, WANs, network segments and Internet systems in a three-tier data center architectural design. Review and determine current network and system requirements, used as a basis to write recommendations for best practices, risk and security management procedures

Management of Storage Area Network (SAN) hardware, perform routine audits of storage infrastructure using monitoring tools such as HP OpenView, HP Systems Insight Manager. Coordination of data center activities and operations

Securing network systems by enforcing policies and monitoring access using intrusion detection systems Wireshark, SNMP, Syslog

Configure, Generate, manage, and leverage the use of SSH key pairs to authenticate to remote systems including enterprise servers and network devices

This enabled seamless remote administration, and encrypted file transfer using SSH in conjunction with other protocols such as SFTP, SCP as well as application integration, eliminating the need to provide passwords for authentication.

Maintain awareness of network activities, upgrades, and maintenance of the network, perform research, and write technical documentation to improve service levels. Evaluate and modify system's performance based on specific needs and requirements. Identify, control, and track configuration items (CI) and any changes made to them in a configuration management database (CMDB)

Troubleshooting network/Storage hardware problems and outages; scheduling upgrades; collaborating with network architects and other Network Engineers on network optimization

Determine and design IP Addressing Schemes, configure VLANs and VLAN Trunks using 802.1Q protocol, configure Telnet, SSH for remote device management. Configure port security

Responsible for monitoring critical network elements and applications using various Network Monitoring Systems in the 24 x 7 x 365 Engineering Private Cloud Environment, Managing, configuration and deployment of virtualization solutions

Vulnerability Management – Research and remediate vulnerabilities mostly related to Operating Systems and applications

Configure HP Servers: HP ProLiant DL580 Gen7 Servers, HP ProLiant DL580 Gen5 Servers, HP ProLiant DL380p Gen8 Servers and HP ProLiant BL460c Gen7 Blade Servers, HP Z400, Z420, Z820 Workstation

Configure Integrated Lights Out (ILO) IP Addresses for remote Server management/access

Install and configure NVIDIA Quadro K6000 graphics cards, DataOn Host Bus Adapters (HBA) in Server Systems

Install DataOn Storage Systems (Enclosures)/JBOD and Configure RAID Arrays/Windows Storage Spaces

Install, Configure and Manage a Datacenter/Private Cloud backup Infrastructure using System Center Data Protection Manager 2012 R2

Add users to the Engineering domain /Active Directory, reset passwords, configure Group Policy objects, and create new computer accounts. Manage Multiple Active directory sites

Image Computers and perform installation of system images using WDS (Zero touch and the Lite Touch) Installation
Microsoft Team Foundation Server builds process implementation, troubleshooting and Administration
Management of TFS/SQL Servers – Configuration, troubleshooting, performance monitoring, upgrades, and enhancements. Management, Maintenance & Administration of MS project Server 2010 and SharePoint Enterprise.
Create New Projects and SharePoint sites for the User Community, Handle customer issues pertaining to service delivery, Participate in Review meetings to identify improvement Plans
Investigate and troubleshoot high priority issues and support Incident Management process
Installation, Configuration and Management of Windows Server 2008R2/2012R2 Hyper-V Hosts, Failover clusters, SCCM/SCVMM, Network File share Provisioning and NTFS permissions. Directly responsible for mitigating any technical failures within and about the Private Windows Cloud Environment
Windows Server Administration including providing solutions for (Windows Server 2000, 2003, 2008R2 & 2012R2), Active Directory Administration, DNS and DHCP and Proxy Servers
Install, maintain, administer, and troubleshoot Desktop Operating Systems (Windows XP, Windows 7, and Windows 8) on physical machines as well as on virtual systems
Management and maintenance of License Servers (for Cosmic, Wind River, Install Shield, Coverity licenses)
Co-ordinate with multiple technical teams, Business users, Customers, hardware & Software vendors as well as offshore teams, generally responsible for coordinating the onsite-offshore operation
Engage vendors for hardware/Software problems covered under existing support contracts, accept incoming shipments and track server assets through deployment. Execute hardware Lifecycle Management plans
Configuration and Management of Windows Server Update Server (WSUS) for multiple sites
Setup End user computing devices (HP Thin Clients), Dell/HP Desktops and Virtual machines for new users Troubleshoot Print device issues, manage and maintain print servers, coordinate/assist with the introduction of new services
Develop and Manage Workflow Process Automation and act as Community Facilitator for the Engineering Workflow Processes Automation.